

Bara Norbert

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EDUCATION

Universitatea Babe-Bolyai

BSc Computer Science

Cluj-Napoca, Romania

2023 – Present

Colegiul Naional Nagy Mózes

High School Diploma

Târgu Secuiesc, Romania

Graduated 2023

PROJECTS

E-Commerce Hybrid Recommendation System | *Java, Apache Spark, Spring Boot, Redis*

- Engineered a scalable hybrid recommendation engine integrating Alternating Least Squares (ALS) collaborative filtering and GRU neural networks for session-based modeling
- Developed a price-adaptive re-ranking layer to align recommendations with user affordability constraints, significantly increasing NDCG and catalog recall
- Architected a containerized data flow, utilizing Spark for batch processing and a Spring Boot REST API backed by PostgreSQL and Redis for low-latency delivery

Fixar Marketplace Platform | *Next.js 14, Fastify, Prisma, PostgreSQL, Socket.io*

In Progress

- Developing a dual-language tradesperson marketplace utilizing Next.js App Router for SEO-optimized SSR and a Fastify backend
- Architected an ad-style listing system for tradespeople, managing relational data and queries with Prisma and PostgreSQL
- Implemented real-time bidirectional chat messaging between clients and tradespeople utilizing Socket.io

Events Manager | *Vue 3, TypeScript, Django REST API, PostgreSQL, Redis*

- Developed a scalable, production-ready web application for event management with a Vue frontend and Django backend
- Engineered advanced features including real-time updates via WebSockets and chunked file uploads for large media
- Implemented secure user authentication with role-based access control and an automated security monitoring background thread

Hospital Management Application | *C#, WinUI, MVVM Architecture*

- Built a comprehensive desktop application utilizing C# and the MVVM architectural pattern
- Collaborated effectively in a team environment to hit defined milestones, successfully implementing complex business logic

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, JavaScript, TypeScript, SQL, Bash

Frontend: Vue, React, Next.js, Tailwind CSS, HTML

Backend & Databases: Spring Boot, Django, Fastify, ASP.NET Core, PostgreSQL, Redis, MongoDB

Tools & DevOps: Apache Spark, Git, Docker, Prisma, Vercel, Unix Shell

AWARDS & LANGUAGES

Awards: 2nd Prize in OTI 2023 (C#), Participated in numerous regional mathematics and programming competitions

Languages: Hungarian (Native), English (Cambridge C1 Certified)